ALL INDIA TEST SERIES

FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FULL TEST – V

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

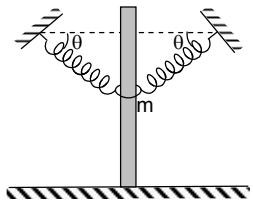
PART – I

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

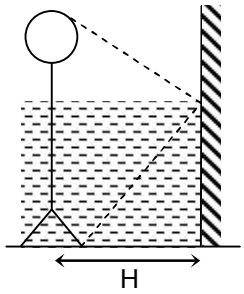
1. A ring of mass m can freely slide on a smooth vertical rod. The ring is symmetrically attached with two springs (as shown in the figure). Each of spring constant k. Each spring makes an angle θ with the horizontal. If the ring is slightly displaced vertically, its time period is:



(A) $2\pi\sqrt{\frac{m}{2k}}$ (B) $2\pi\sqrt{\frac{m}{k}}$

(C) $2\pi\sqrt{\frac{2m}{k\sin^2\theta}}$ (D) $2\pi\sqrt{\frac{m}{2k\sin^2\theta}}$.

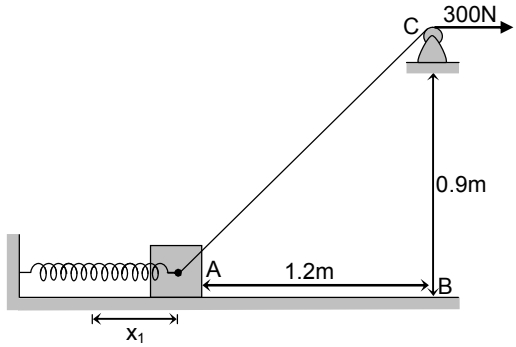
2. A man of height H stands in front of a plane mirror. Water has been filled upto a height 3H/4. The man can just see his foot in the mirror placed parallel in front of him. What is the refractive index of water?



(A) $\frac{20}{3\sqrt{17}}$ (B) $\frac{5}{3\sqrt{17}}$

(C) $\frac{4}{3\sqrt{17}}$ (D) $\frac{5}{\sqrt{17}}$

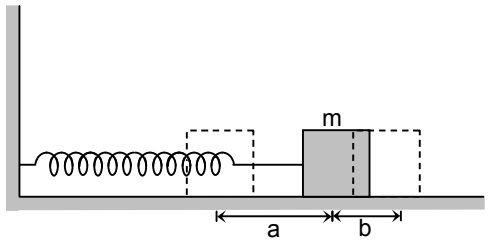
3. A 590 kg block at A moves along the horizontal surface with negligible friction under the action of the constant 300 N force in the cable. The block is released from rest at A, with the spring to which it is attached extended an initial amount x_1 = 0.3 m. The spring has a stiffness constant, K = 80 N/m. The kinetic energy of the block as it reaches position B is:



(A) 86.4 J
(B) 180 J
(C) 93.6 J
(D) None of these

Space for Rough work

4. A mass-spring system as shown in the figure oscillates such that the mass moves on a rough surface having coefficient of friction μ. It is compressed by a distance a from its normal length and, on being released. It moves to a distance b from its equilibrium position. The decrease in amplitude for one half-cycle (–a to b) is:

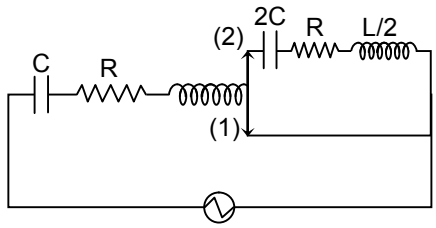


(A) $\frac{\mu mg}{K}$ (B) $\frac{2\mu mg}{K}$

(C) $\frac{\mu g}{K}$ (D) None of these

5. In a series LR circuit, the frequency of the AC source increases. Which of the following options is correct?
(A) The phase difference between the current and the voltage increases.
(B) The power factor increases.
(C) The amplitude of the current increases.
(D) The amplitude of potential drop across the inductor decreases.

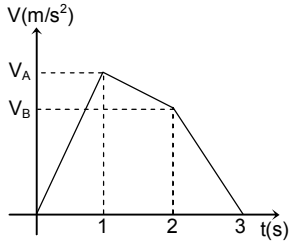
6. In the diagram shown, when the switch is at position 1 then the frequency of maximum current from the battery is ω. What will be the frequency of maximum current from the battery when the switch is at 2?



(A) ω
(B) 2ω
(C) $\omega/\sqrt{3}$
(D) None of these

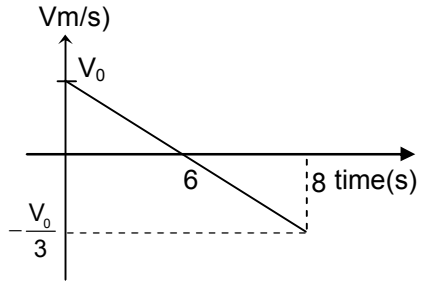
Space for Rough work

7. Consider the velocity versus time graph of particle moving on a straight line. The average velocity on the 1st one second is V_1m/s and the average velocity on the 2nd one second is V_2 m/s.



What is the average velocity on the last one second?
(A) $V_2 - V_1$
(B) $(V_2 + V_1)/2$
(C) $2V_2$
(D) None of these

8. The velocity versus time graph of a particle is given.



If the total displacement of the particle is 4m then what is the initial velocity of the particle?
(A) 1.0 m/s
(B) 1.1 m/s
(C) 1.5 m/s
(D) None of these

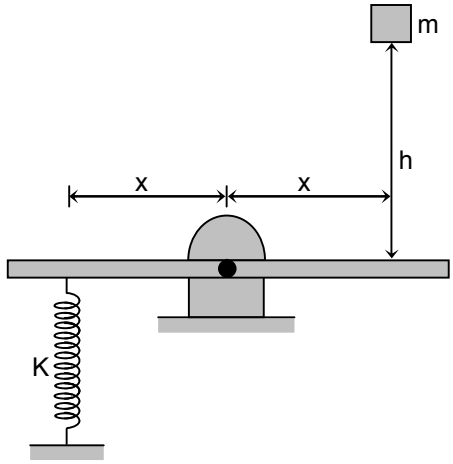
9. The potential energy of harmonic oscillator of mass 2 kg in its mean position is 5J. If its total energy is 9J and its amplitude is 0.01m, its time period will be:

(A) $\frac{\pi}{100}$s (B) $\frac{\pi}{50}$s

(C) $\frac{\pi}{20}$s (D) $\frac{\pi}{10}$s

Space for Rough work

10. A crate of mass m falls from a height h onto the end of a platform, as shown in the figure. The spring is initially unstretched and the mass of the platform can be neglected. Assuming that there is no loss of energy, the maximum elongation of the spring is:



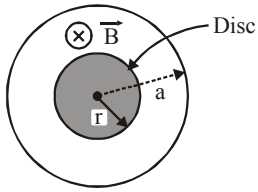
(A) $\dfrac{mg+\sqrt{m^2g^2+2mghK}}{K}$

(B) $\dfrac{mg-\sqrt{m^2g^2+2mghK}}{K}$

(C) $\dfrac{K}{\sqrt{m^2g^2+2mghK}}$

(D) None of these

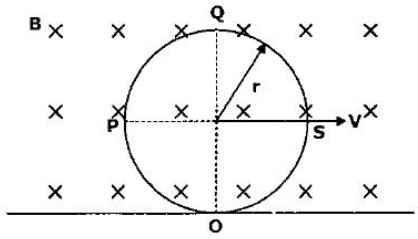
11. A uniform disc of radius r and mass m is charged uniformly with the charge q. This disc is placed flat on a rough horizontal surface having coefficient of friction μ. A uniform magnetic field is present in a circular region of radius a (> r) but varying as kt^3 as shown in figure. Find the time in second after which the disc begins to rotate. (Given r=1 m, m=18 kg, q=1C, μ=0.1, K=4, g=10 m/s^2)



(A) 2 (B) 4

(C) 8 (D) None of these

12. A conducting ring of radius r and resistance R rolls on a horizontal surface with constant velocity v. The magnetic field B is uniform normal to the plane of the loop.



(A) The induced emf, between O and Q is 2Bvr.

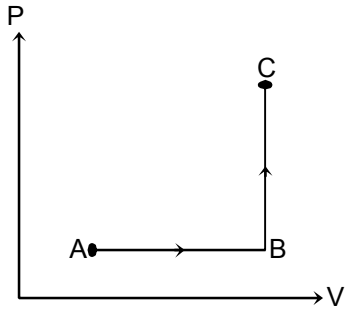
(B) An induced current $I=\dfrac{2Bvr}{R}$ flows in the clockwise direction.

(C) An induced current $I=\dfrac{2Bvr}{R}$ flows in the anticlockwise direction.

(D) None of these

Space for Rough work

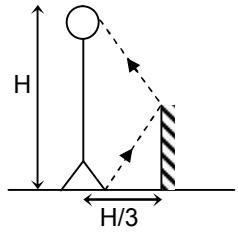
13. The P-V diagram of a system undergoing thermodynamic transformation as shown in the figure. The work done by the system in going from A $\rightarrow$ B $\rightarrow$ C is 30J and 4J heat is given to the system. The change in internal energy between A and C is
(A) 26 J
(B) – 26 J
(C) 34 J
(D) – 34 J



14. A sample of gas expands from volume V_1 to V_2. The amount of work done by the gas is greatest when the expansion is:
(A) Isothermal
(B) Isobaric
(C) Adiabatic
(D) Equal in all cases.

15. In a given process for an ideal gas, dW = 0 and dQ < 0. Then for the gas:
(A) The temperature will decrease
(B) The volume will increase
(C) The pressure will remain constant
(D) The temperature will increase.

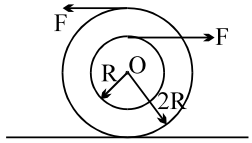
16. In the diagram shown, the length of the mirror is half of the height of the man. What length of his image can the man see in the mirror?
(A) H/3
(B) H/2
(C) H
(D) zero



17. Starting from rest at the same time, a coin and a ring roll down an incline without slipping. Which reaches the bottom of the incline first?
(A) The ring reaches the bottom first
(B) The coin reaches the bottom first
(C) They arrive at the bottom simultaneously
(D) The winner depends on the relative masses of the two

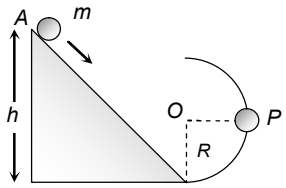
Space for Rough work

18. In the given figure F = 10 N, R = 1 m, mass of the body is 2 kg and moment of inertia of the body about an axis passing through O and perpendicular to plane of body 4kg-m^2. O is the centre of mass of the body. If the ground is smooth what is total kinetic energy of the body after 2 seconds



(A) 100 J (B) 75 J
(C) 50 J (D) 25 J

19. A solid ball of mass *m* and radius *r* rolls without slipping along the track shown in the fig. The radius of the circular part of the track is *R*. The ball starts rolling down the track from rest from a height of 8*R* from the ground level. When the ball reaches the point P then its velocity will be



(A) $\sqrt{gR}$ (B) $\sqrt{5gR}$
(C) $\sqrt{10gR}$ (D) $\sqrt{3gR}$

20. Three measurements are made as 18.425 cm, 7.21 cm and 5.00 cm. The addition should be written as:
(A) 30.635 cm (B) 30.64 cm
(C) 30.63 cm (D) 30.6 cm

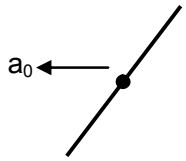
21. Two identical charged particles enter in a uniform magnetic field with same speed but at angles 30° and 60° with field. Let a, b and c be the ratio of their time periods, radii and pitches of the helical paths then:
(A) abc = 1 (B) abc > 1
(C) abc < 1 (D) ab = c

22. The acceleration due to gravity on the surface of the moon is $\frac{1}{6}$ that on the surface of earth and the diameter of the moon is one-fourth that of earth. The ratio of escape velocities on earth and moon will be:
(A) $\frac{\sqrt{6}}{2}$ (B) $\sqrt{24}$
(C) 3 (D) $\frac{\sqrt{3}}{2}$

Space for Rough work

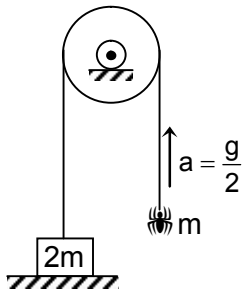
23. The greatest angle through which a man can oscillate on a swing the ropes of which can support twice his weight when at rest is:
(A) 90° (B) 60°
(C) 120° (D) 30°

24. A bead of mass m is fitted on a rod and can move on it without friction. Initially the bead is at the middle of the rod and the rod moves translationally in a horizontal plane with an acceleration a_0 in horizontal direction forming angle α with the rod. The acceleration of bead with respect to rod is:



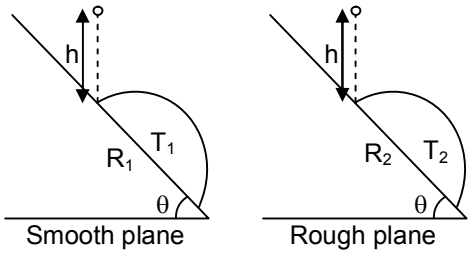
(A) $g \sin \alpha$ (B) $(g + a_0) \sin \alpha$
(C) $g \sin \alpha + a_0 \cos \alpha$ (D) $g \sin \alpha - a_0 \cos \alpha$

25. An insect of mass m crawls along the hanging thread with an acceleration $a = \frac{g}{2}$. The reaction offered by ground on the block of mass 2m is:



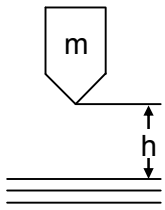
(A) $\frac{3mg}{2}$ (B) mg
(C) 2mg (D) $\frac{mg}{2}$

26. Two balls are dropped from same height h on a smooth plane and the other on a rough plane having same inclination θ with horizontal. Both the planes have same coefficient of restitution. If range and time of flight of first and second balls are R_1, T_1 and R_2, T_2 respectively, then:



(A) $T_1 = T_2$, $R_1 = R_2$
(B) $T_1 < T_2$, $R_1 < R_2$
(C) $T_1 = T_2$, $R_1 > R_2$
(D) $T_1 > T_2$, $R_1 > R_2$

27. An iron nail is falling from a height h. If it penetrates through a distance x into the sand, the average resistance of the sand is:



(A) $mg\frac{h}{x}$ (B) $mg\left(\frac{h}{x}+1\right)$
(C) $mg\frac{x}{h}$ (D) mg

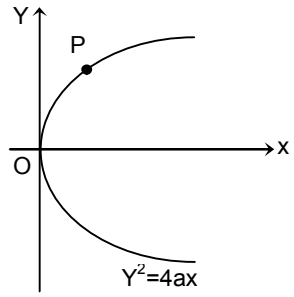
Space for Rough work

28. A ray of light traveling in a medium of refractive index n_1 is incident on a medium of refractive index $n_2 (< n_1)$ at an angle of incidence 'θ'. The reflected and refracted rays make an angle of 120° with each other. Then the critical angle for the boundary separating the two media is:

(A) $\sin^{-1}\left(\frac{1}{2}\tan\theta\right)$ (B) $\sin^{-1}\left(\frac{1}{\sqrt{3}+\tan\theta}\right)$

(C) $\sin^{-1}\left(\frac{2\tan\theta}{\sqrt{3}-\tan\theta}\right)$ (D) $\sin^{-1}(2\cot\theta)$

29. A thin wire is bent into the shape of a parabola in the vertical plane as shown. A small bead 'P' can freely slide along the wire.
The coefficient of friction between the bead and the wire is 'μ'.
The condition that bead does not slip on the wire is



(A) $y \geq 2a\mu$ (B) $y \geq \frac{2a}{\mu}$

(C) $y \leq 2a\mu$ (D) $y \leq \frac{2a}{\mu}$.

30. In a nuclear reactor, energy is produced by the following process :

$${}_1H^2 + {}_1H^2 \longrightarrow {}_1H^3 + P$$

$${}_1H^3 + {}_1H^2 \longrightarrow {}_2He^4 + n$$

The reactor produces a power of 10^{10} MW with 50% efficiency. Then the number of deutrons required to run the reactor for one year will be approximately :

$m({}_1H^2) = 2.014\,u,\ m(p) = 1.007\,u,\ m(n) = 1.008\,u,\ m({}_2He^4) = 4.001u$

(A) 5×10^{35} (B) 5×10^{32}

(C) 5×10^{38} (D) 5×10^{40}

Space for Rough work

Chemistry

PART – II

SECTION – A

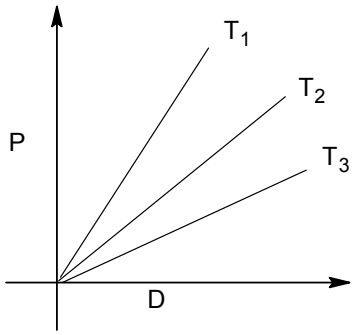
Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Lattice energies of BeF_2, MgF_2, CaF_2 and BaF_2 are –2906, –2610, –2459 and –2367 KJ/mol respectively. Hydration energies of Be^{2+}, Mg^{2+}, Ca^{2+}, Ba^{2+} and F^- are –2494, –1921, –1577, –1305 and –457 KJ mol^{-1} respectively. Which of the fluorides is not soluble in water?
   (A) BeF_2 (B) MgF_2
   (C) CaF_2 (D) BaF_2

2. The stability of dihalides of Si, Ge, Sn and Pb increases steadily in the sequence:
   (A) $GeX_2 < SiX_2 < SnX_2 < PbX_2$ (B) $SiX_2 < GeX_2 < PbX_2 < SnX_2$
   (C) $SiX_2 < GeX_2 < SnX_2 < PbX_2$ (D) $PbX_2 < SnX_2 < GeX_2 < SiX_2$

3. Pressure Vs. density curve for an ideal gas at three different temperatures T_1, T_2 and T_3 is shown below, which is correct relation here?

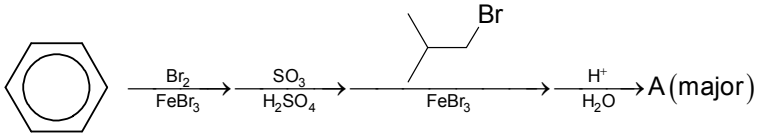


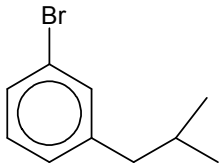
   Which of these is correct?
   (A) $T_1 > T_2 > T_3$ (B) $T_1 > T_3 > T_2$
   (C) $T_3 > T_2 > T_1$ (D) Can't be said

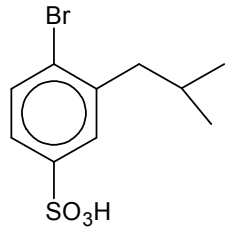
4. A metal oxide has the formula Z_2O_3. It can be reduced by hydrogen to give free metal and water. 0.16 g of the metal oxide requires 6 mg of hydrogen for complete reduction. The atomic weight of metal is
   (A) 27.9 (B) 159.6
   (C) 79.8 (D) 55.8

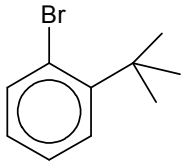
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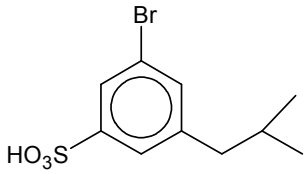
5. Give the major product from the following reaction sequence:

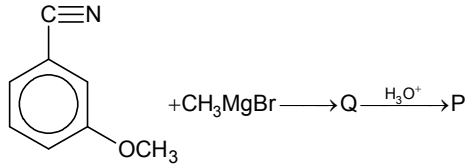


(A) 

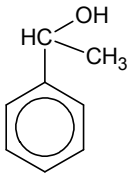
(B) 

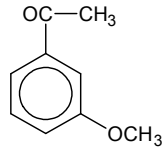
(C) 

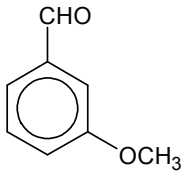
(D) 

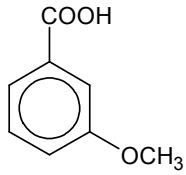
6. 

The product 'P' in the above reaction is

(A) 

(B) 

(C) 

(D) 

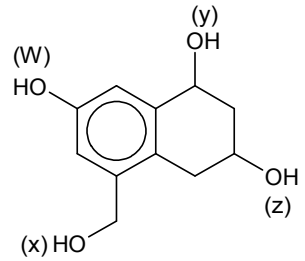
7. In an open-end manometer, one end filled with mercury is attached to a gas-filled container and the other end is open to the atmosphere. If the gas pressure in the container is less than atmospheric pressure.
   (A) The Hg level will be higher in the arm open to the atmosphere
   (B) Hg will be forced out of the open and of the U-tube.
   (C) The difference between the Hg levels in the two arms will be greater than 76 cm.
   (D) The Hg level will be higher in the arm connected to the container.

Space for Rough work

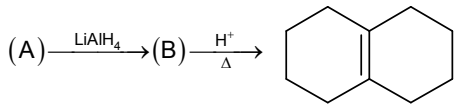
8. 100 ml of $NaHC_2O_4$ required 50 ml of 0.1 M $KMnO_4$ solution in acidic medium. Volume of 0.1 M NaOH required by 100 ml of $NaHC_2O_4$ is:
(A) 50 ml (B) 100 ml
(C) 125 ml (D) 150 ml

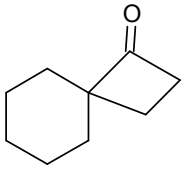
9. Ratio of σ and π bonds is maximum in
(A) Naphthalene (B) Tetra Cyanomethane
(C) Enolic form of urea (D) Equal in all

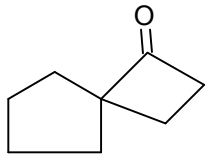
10. Which of the following statement is true for $CsBr_3$?
(A) It is a covalent compound (B) It contains Cs^{3+} and Br^{-1}
(C) It contains Cs^{+} and Br_3^{-} (D) It contains Cs^{+}, Br^{-} and lattice Br^2 molecule

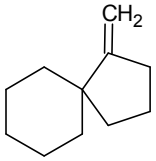
11. 

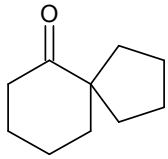
Decreasing order of acidic strength of different (- OH) group is:
(A) w > x > y > z (B) w > z > x > y
(C) z > w > x > y (D) z > x > w > y

12. 

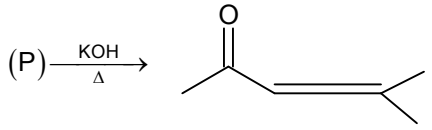
What is (A)?
(A) 

(B) 

(C) 

(D) 

Space for Rough work

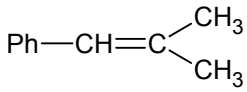
13. 

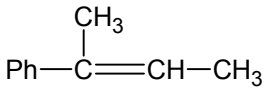
$(Q) \xrightarrow[\Delta]{KOH}$ $Ph—CH_2—OH + Ph—COO^{\ominus}K$

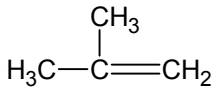
$(R) \xrightarrow{O_3} P + Q$

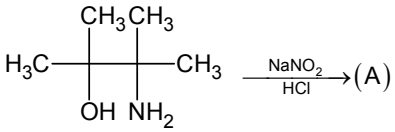
What is the structure of (R):

(A) $Ph—CH=CH—CH_3$

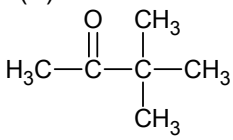
(B) 

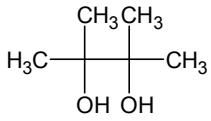
(C) 

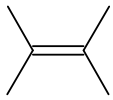
(D) 

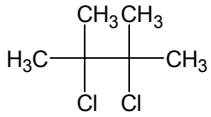
14. 

Product (A) is:

(A) 

(B) 

(C) 

(D) 

15. HNO_3 in pure state is colorless but it is often of yellow color. It is due to

(A) Unstable structure of HNO_3 which immediately changes to NO_2^-

(B) Photochemical decomposition in presence of sunlight

(C) Interaction of atmospheric gases with HNO_3

(D) Conversion of HNO_3 into NO_2^+

16. When K_2CrO_4 is added to $CuSO_4$ solution, there is formation of $CuCrO_4$ as well as $CuCr_2O_7$. Formation of $CuCr_2O_7$ is due to

(A) Basic nature of $CuSO_4$ solution which converts CrO_4^{2-} to $Cr_2O_7^{2-}$

(B) Acidic nature of $CuSO_4$ solution which converts CrO_4^{2-} to $Cr_2O_7^{2-}$

(C) $CuSO_4$ has the typical property of converting $CuCrO_4$ formed to $CuCr_2O_7$

(D) 10 $CuCr_2O_7$ is formed

Space for Rough work

17. A large excess of $MgF_2(s)$ is maintained in contact with 1 lit of pure water to produce a saturated solution. When an additional 1 lit. of pure water is added to the solid-liquid mixture and equilibrium re-established, compared to its value in the original saturated solution $\left[Mg^{2+}\right]$ will be:

(A) The same
(B) twice as large
(C) three times as large
(D) half as large

18. A mixture of 2 colourless substances was dissolved in water. When gaseous Cl_2 was passed through the solution, a deep brown color developed. Addition of $BaCl_2$ or NaOH to the original solution give a white precipitate. The mixture contained.

(A) $NaNO_3$ and $AlCl_3$
(B) NaBr and KCl
(C) MgI_2 and $MgSoO_4$
(D) $BaSO_4$ and $PbCl_2$

19. $(Ag+Pb)\text{alloy} \xrightarrow[\text{add zinc}]{\text{melt and}} (Ag+Pb+Zn)\text{melt} \xrightarrow{\text{Cool}} \frac{\text{layer X}}{\text{layer Y}}$

Select correct statement based on above scheme.

(A) Layer X contain zinc and silver
(B) Layer Y contains lead and silver but amount of silver in this layer is smaller than in the layer X
(C) 'X' and 'Y' are immisible liquids
(D) All are correct

20. What is the molarity of HCl in a solution prepared by dissolving 5.5 g HCl in 200 g ethanol if the density of the solution is 0.79 g/ml?

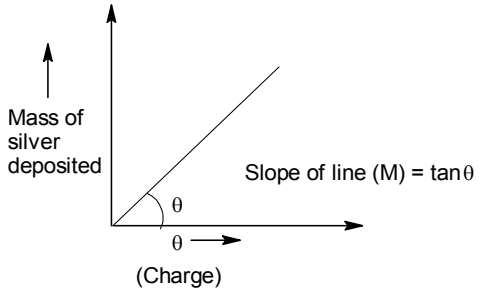
(A) 2.1 M
(B) 0.93 M
(C) 1.7 M
(D) 0.58 M

21. Select the incorrect statement among the folloing:

(A) Nearest neighbour distance in NaCl = a/2
(B) Nearest neighbour distance in $CaF_2 = \frac{a\sqrt{3}}{4}$
(C) Nearest neighbour distance in CsCl = $\frac{a}{2}$
(D) Nearest neighbour distance in Na_2O = $\frac{a\sqrt{3}}{4}$

Space for Rough work

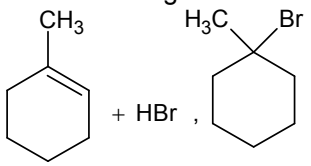
22. In the electrolysis of $AgNO_3$, the mass of silver deposited is plotted against the charge:

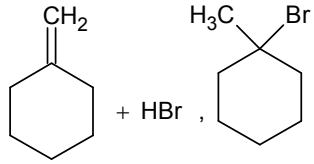


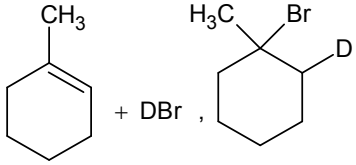
Slope of the line gives:

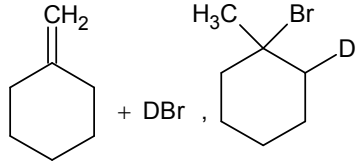
(A) The equivalent mass of silver
(B) Electrochemical equivalent of silver
(C) the value of Faraday
(D) the current passed through the cell

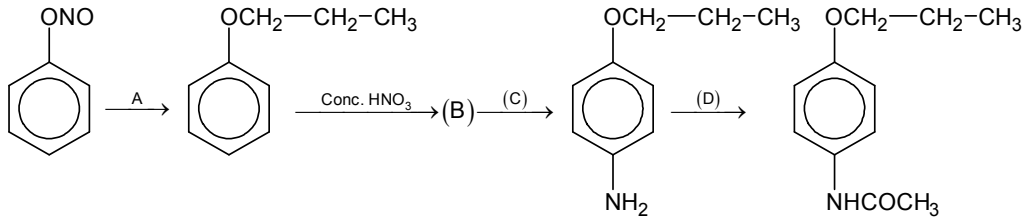
23. Which of the following is not correctly matched?

(A) 

(B) 

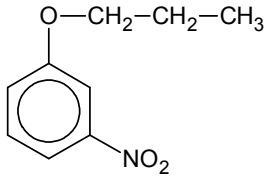
(C) 

(D) 

24. 

In the given reaction

(I) A is $CH_3-CH_2-CH_2-Br$

(II) B is 

(III) C is Sn/HCl

(IV) D is $(CH_3CO)_2O$

Out of these which are correct?

(A) (I) and (II)
(B) (I) and (III)
(C) (I), (III) and (IV)
(D) (I), (II), (III) and (IV)

Space for Rough work

25. Match List-I and List – II and select the correct answer using the codes given below the lists:

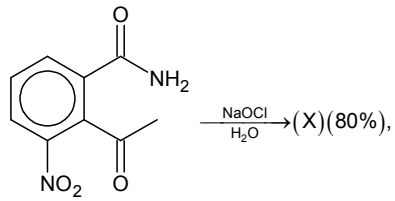
(a)	α-D and β-D glucose	(1)	Enantiomers
(b)	D-glucose and D-galactose	(2)	Anomers
(c)	Erythose and threose	(3)	Epimers
(d)	D-glyceraldehyde and L-glyceraldehyde	(4)	Diasteromers

(A) (a $\rightarrow$ 3), (b $\rightarrow$ 2), (c $\rightarrow$ 1), (d $\rightarrow$ 4) (B) (a $\rightarrow$ 2), (b $\rightarrow$ 3), (c $\rightarrow$ 4), (d $\rightarrow$ 1)
(C) (a $\rightarrow$ 3), (b $\rightarrow$ 2), (c $\rightarrow$ 4), (d $\rightarrow$ 1) (D) (a $\rightarrow$ 2), (b $\rightarrow$ 3), (c $\rightarrow$ 1), (d $\rightarrow$ 4)

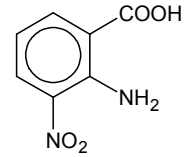
26. Which of the following statements is wrong for inter halogen?
(A) The value of n is AX_n (inter halogen) can be 1, 3, 5 or 7
(B) The value of n is AX_n (inter halogen) can be 2, 4 or 6
(C) A can never be fluorine as it is most electronegative halogen
(D) X can never be iodine as it is least electronegative halogen

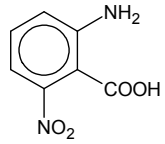
27. What are the products formed in the reaction of Xenon hexafluoride with silica?
(A) $XeSiO_4 + HF$ (B) $XeF_2 + SiF_4$
(C) $XeOF_4 + SiF_4$ (D) $XeO_3 + SiF_4$

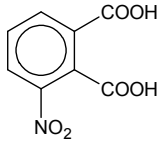
28. The straight chain polymer is formed
(A) hydrolysis is CH_3SiCl_3 followed by condensation polymerisation
(B) hydrolysis of $(CH_3)_4Si$ followed by addition polymerisation
(C) hydrolysis of $(CH_3)_2SiCl_2$ followed by condensation polymerisation
(D) hydrolysis of $(CH_3)_3SiCl$ followed by condensation polymerisation

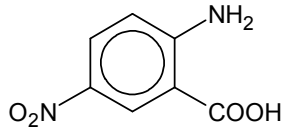
29. 

Product X will be

(A) 

(B) 

(C) 

(D) 

30. Which of the following is a covalent carbide?
(A) CaC_2 (B) Al_4C_3
(C) WC (D) SiC

Space for Rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. The range of function f(x) where $f(x) + 2f(1 - x) = 1 - x^2$, $\forall\ x \in R$ is
   (A) $(-\infty, 1]$ (B) $[1, \infty)$
   (C) $(-1, 1)$ (D) $[-1, 1]$

2. Six digit numbers are formed using 1, 2, 3, 4, 5, 6, (repetition is not allowed). The number of ways such that even digits are in increasing order and odd digits are in decreasing order, is
   (A) 20 (B) 40
   (C) 60 (D) 120

3. The average of $2 \sin 2^\circ, 6 \sin 6^\circ, 10 \sin 10^\circ, \ldots\ldots 178 \sin 178^\circ$ is
   (A) $\cot 1^\circ$ (B) $2 \operatorname{cosec} 2^\circ$
   (C) $2(1 + \tan 2^\circ)$ (D) $2(1 + 2 \tan 2^\circ \sin 44^\circ)$

4. The number of points with integral co-ordinates that lie in the interior of the region common to the circle $x^2 + y^2 = 16$ and the parabola $y^2 = 4x$ is
   (A) 10 (B) 16
   (C) 17 (D) 13

5. The projection of the line segment joining the points (1, –2, 3) and (3, 1, 0) on the line whose direction ratios are (2, 1, 3) is
   (A) $\frac{1}{\sqrt{14}}$ (B) $\frac{2}{\sqrt{14}}$
   (C) $\frac{3}{\sqrt{14}}$ (D) $\frac{5}{\sqrt{14}}$

Space for rough work

6. Set A has 6 elements and its power set has n elements. Let m be the sum of distinct cardinal numbers of elements of power set of A, then the value of n – m is
(A) 43 (B) 40
(C) 35 (D) 50

7. Number of solutions of the equation cos 9x cos 3x = cos 18x cos 12x, $x \in [0, \pi]$ is
(A) 12 (B) 30
(C) 28 (D) 32

8. If the roots of the equation $x^2 + mx + n = 0$ are cubes of the roots of the equation $x^2 + px + q = 0$, then
(A) $m = p^3 + 3pq$ (B) $m = p^3 - 3pq$
(C) $m + n = p^3$ (D) $\frac{m}{n} = \left(\frac{p}{q}\right)^3$

9. The equation of the hyperbola whose foci are symmetrically situated on the x-axis with respect to the origin, the equation of the asymptotes as $y = \pm\frac{4}{3}x$ and distance between foci being 20, is
(A) $\frac{x^2}{64} - \frac{y^2}{36} = 1$ (B) $\frac{x^2}{16} - \frac{y^2}{9} = 1$
(C) $\frac{x^2}{9} - \frac{y^2}{16} = 1$ (D) $\frac{x^2}{36} - \frac{y^2}{64} = 1$

10. The value of $\int_0^{\infty} \frac{\ln\left(x + \frac{1}{x}\right)}{\left(1 + x^2\right)} dx$ is
(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{2}\ln 2$
(C) $\frac{\pi}{2}\ln 4$ (D) none of these

11. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 4\hat{j} - \hat{k}$, $\vec{c} = \hat{i} + \hat{j} + 2\hat{k}$ and $\hat{s}$ be unit vector the magnitude of vector $(\vec{a}\cdot\hat{s})(\vec{b}\times\vec{c}) + (\vec{b}\cdot\hat{s})(\vec{c}\times\vec{a}) + (\vec{c}\cdot\hat{s})(\vec{a}\times\vec{b})$ is equal to
(A) 1 (B) 2
(C) 3 (D) 4

Space for rough work

12. The number of solutions of the equation $|x - 1| + |x - 10| + x = 9 + [x]$ is, (where [.] denote the greatest integer function)
(A) 10 (B) 2
(C) 4 (D) 20

13. Two dice are thrown at a time, the probability that the difference of the numbers is 2, is
(A) $\frac{1}{9}$ (B) $\frac{2}{9}$
(C) $\frac{4}{9}$ (D) $\frac{5}{9}$

14. The four sides of a quadrilateral are given by equation $(xy + 12 - 4x - 4y)^2 = (2x - 2y)^2$. The equation of a line with slope $\sqrt{3}$ which divides the area of quadrilateral in two equal parts is
(A) $y = \sqrt{3}(x + 4)$ (B) $y = \sqrt{3}x + 4$
(C) $y = \sqrt{3}(x - 4) + 4$ (D) $y = \sqrt{3}(x - 4)$

15. Let $I = \int_0^1 \frac{1}{\sqrt{1+x^5}} dx$ then which of following is correct
(A) $I < \frac{\pi}{4}$ (B) $I > 1$
(C) $I > 3$ (D) none of these

16. The arithmetic mean of a set of observation is $\bar{X}$. If each observation is divided by α and then is increased by 10, then the mean of the new series is
(A) $\frac{\bar{X}}{\alpha}$ (B) $\frac{\bar{X} + 10}{\alpha}$
(C) $\frac{\bar{X} + 10\alpha}{\alpha}$ (D) $\alpha\bar{X}$ + 10

17. Let $w \neq 1$ and $w^{13} = 1$. If $a = w + w^3 + w^4 + w^{-4} + w^{-3} + w^{-1}$ and $b = w^2 + w^5 + w^6 + w^{-6} + w^{-5} + w^{-2}$, then the quadratic equation whose roots are a and b is
(A) $x^2 + x + 3 = 0$ (B) $x^2 + x + 4 = 0$
(C) $x^2 + x - 3 = 0$ (D) $x^2 + x - 4 = 0$

Space for rough work

18. On the ellipse $\frac{x^2}{18}+\frac{y^2}{8}=1$, the point M nearest to the line 2x – 3y + 25 = 0 is

(A) (–3, 2) (B) $\left(-\sqrt{2}, \frac{8}{3}\right)$

(C) (3, 2) (D) $\left(-3\sqrt{2}, 0\right)$

19. The value of $\lim\limits_{x\to\infty}\left(\sqrt[4]{x^4+3x^3}-\sqrt[3]{x^3+2x^2}\right)$ is

(A) $\frac{1}{6}$ (B) $\frac{1}{18}$

(C) $\frac{1}{12}$ (D) $\frac{1}{24}$

20. In any triangle ABC, a, b, c be the sides opposite to angles A, B and C respectively and b + c = 3a. Then the value of $\left[\cot\frac{B}{2}\cot\frac{C}{2}+\tan\frac{B}{2}\tan\frac{C}{2}\right]$ is, (where [.] denotes the greatest integer function)

(A) 1 (B) 2

(C) 3 (D) 4

21. Number of solutions of the equation $\sin^{-1}(\sin x)+\cos^{-1}(\cos x)=x$, $\forall\, x \in (0, \pi]$ are

(A) 0 (B) 1

(C) 2 (D) 3

22. Let $f(x)=x^3+x^2+2x-1$. The minimum value of [x] that satisfy the f(f(x)) > f (2x + 1) is _____. (Where [.] denotes the greatest integer function).

(A) 0 (B) 1

(C) 2 (D) 3

23. If the two feet of normals drawn from a point to the parabola $x^2 – 6x – 4y + 5 = 0$ be (7, 3) and (–1, 3), then the third foot is

(A) (1, 0) (B) (–3, 8)

(C) (3, –1) (D) (5, 0)

Space for rough work

24. If α, β, γ are number of elements in sets A, B and C respectively and sum of number of all subsets of A, B and C is 28, then maximum number of ordered triplets (α, β, γ) is
(A) 1 (B) 2
(C) 4 (D) 6

25. A flag staff of 5 m height stands on a building 25 m high. At an observer at a height of 30 m, the flag staff and the building subtend equal angles. The distance of the observer from the top of flag staff is
(A) $\frac{5\sqrt{3}}{2}$ (B) $5\sqrt{\frac{3}{2}}$
(C) $\frac{2}{3}$ (D) None of these

26. Sum of series $\frac{1}{1.3}+\frac{1}{3.5}+\frac{1}{5.7}+.....$ upto 10 terms is
(A) $\frac{10}{21}$ (B) $\frac{1}{2}$
(C) $\frac{11}{21}$ (D) $\frac{9}{20}$

27. Solution of the differential equation $\left(x^2+6y\right)\frac{dy}{dx}+2\left(xy+1\right)=0$ is
(A) $yx^2 + 2x + 3y^2 = c$ (B) $xy^2 + 2x + 3y = c$
(C) $yx^2 - 2x - 3y^2 = c$ (D) none of these

28. The value of (1 – cot 3º)(1 – cot 7º)(1 – cot 11º)(1 – cot 34º)(1 – cot 38º)(1 – cot 42º) is equal to
(A) 3 (B) 2
(C) 27 (D) 8

29. An incident ray (L_1) is reflected by the mirror (L_2) 3x + 4y = 5 and the equation of reflected ray (L_3) is x + y = 1, the equation of L_1 is
(A) x – y = –3 (B) 17x + 31y = 45
(C) 17x – 31y = –77 (D) none of these

30. If m is a natural number, then m(m + 1)(m + 2) + (7^m – 1) is always divisible by
(A) 5 (B) 9
(C) 6 (D) 12

Space for rough work